

Computational Investigation of the Univalent Bloch Constant: Certified Upper Bounds via Slit-Disk Domains and Barriers to Lower Bound Improvement

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Abstract

The univalent Bloch constant B_u is the infimum of the inradius of the image of the unit disk under normalized univalent functions, and satisfies $0.5708858 < B_u \leq 0.6564$. We conduct a systematic computational investigation of bounds on B_u . Our principal contributions are threefold. First, we establish a *certified* upper bound $B_u \leq 0.6814$ using a three-fold symmetric radial slit-disk domain, computed via an explicit five-step Cayley–Joukowski conformal map chain and verified with 100-digit interval arithmetic. Second, we provide a complete optimization landscape for n -fold radial slit-disk domains ($n = 1, \dots, 12$), confirming that $n = 3$ is optimal among this family. Third, we document in detail why three natural approaches to improving the lower bound — Koebe quarter theorem with Bloch norm dichotomy, the 2×2 Grunsky matrix adversarial search, and higher-fold radial symmetry — all fail to surpass Skinner’s 2009 bound. We analyze the structural barriers to improvement and identify concrete directions for future progress, including SDP hierarchies with large Grunsky matrices, curved-arc domain constructions via the Pólya–Chebotarev problem, and Loewner chain optimal control.

Keywords: Univalent Bloch constant, covering theorems, conformal mapping, slit-disk domains, interval arithmetic, Grunsky inequalities.

MSC 2020: 30C25, 30C55, 30C70, 30C80, 65G20.

1 Introduction

1.1 The univalent Bloch constant

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ denote the open unit disk and let $\mathcal{F}$ be the class of all holomorphic functions $f : \mathbb{D} \rightarrow \mathbb{C}$ normalized by $f'(0) = 1$ [Bhowmik and Sen, 2023]. For $f \in \mathcal{F}$, the *Bloch radius* B_f is the supremum of radii $r > 0$ for which $f(\mathbb{D})$ contains a univalent disk of radius r , i.e., a disk $D(w_0, r) \subset f(\mathbb{D})$ that is the injective image of some subdomain of $\mathbb{D}$.

The **univalent Bloch constant** is defined as

$$B_u := \inf\{B_f : f \in \mathcal{F}, f \text{ is univalent in } \mathbb{D}\}. \quad (1)$$

When f is globally univalent on $\mathbb{D}$, the image $f(\mathbb{D})$ is simply connected and every disk in $f(\mathbb{D})$ is automatically a univalent disk (via f^{-1}). Thus $B_f = \text{inrad}(f(\mathbb{D}))$, the radius of the largest disk contained in $f(\mathbb{D})$, and

$$B_u = \inf\{\text{inrad}(f(\mathbb{D})) : f \in \mathcal{F}, f \text{ univalent}\}. \quad (2)$$

The constant B_u sits atop a hierarchy of four classical constants [Bhowmik and Sen, 2023]:

$$B \leq B_l \leq L \leq B_u, \quad (3)$$

where B is the Bloch constant, B_l the locally univalent Bloch constant, and L the Landau constant. The current best bounds are

$$0.5708858 < B_u \leq 0.6564, \quad (4)$$

with the lower bound due to Skinner [2009] and the upper bound due to Carroll and Ortega-Cerdà [2010]. The gap of approximately 0.086 (relative gap $\approx 14\%$) is the largest among the four constants, suggesting significant room for improvement.

1.2 Motivation and contributions

The univalent Bloch constant has resisted improvement for over fifteen years: Skinner’s lower bound dates from 2009 and Carroll–Ortega-Cerdà’s upper bound from 2008. Understanding *why* the gap remains wide, and identifying concrete paths toward its reduction, is as valuable as any numerical improvement. Our investigation makes the following contributions:

- (i) **Certified upper bound.** We prove $B_u \leq 0.6814202223$ using a three-fold symmetric radial slit-disk domain, with the conformal radius computed via an exact five-step Cayley–Joukowski chain and certified by interval arithmetic at 100-digit precision (Sections 4.1 and 4.4). While weaker than Carroll–Ortega-Cerdà’s 0.6564, our bound is independently derived with full computational rigor.
- (ii) **Complete optimization landscape.** We compute B_f for $n = 1, \dots, 12$ radial slit configurations at optimized inner radii (Section 6.1), confirming that $n = 3$ is optimal among this family and providing a comprehensive picture of the slit-disk approach.
- (iii) **Documented barriers to lower bound improvement.** We show that the Koebe quarter theorem with Bloch norm dichotomy is fundamentally limited by the sharpness of the Koebe constant for the Schlicht class (Section 4.2); that the 2×2 Grunsky matrix constraint is insufficient for univalence enforcement (Section 4.3); and that radial slit domains with $n \geq 4$ fold symmetry produce worse upper bounds (Section 6.1).
- (iv) **Interval arithmetic framework.** We develop a reusable certification pipeline for conformal radius computations, validated by eight comprehensive test suites (Section 4.4).

We emphasize transparency: we did *not* achieve any improvement over the best previously published bounds. The lower bound remains at Skinner’s 0.5708858 and the best upper bound remains at Carroll–Ortega-Cerdà’s 0.6564.

2 Related Work

2.1 Origins of covering theorems

Bloch [1925] proved the foundational result that a universal constant $B > 0$ exists such that the image of any holomorphic function $f: \mathbb{D} \rightarrow \mathbb{C}$ with $f'(0) = 1$ contains a univalent disk of radius at least B . Landau [1929] introduced the Landau constant L and established a systematic framework connecting covering theorems to the theory of normal families and the modular function.

2.2 Upper bounds on B and the Ahlfors–Grunsky conjecture

Ahlfors and Grunsky [1937] established the best known upper bound on the Bloch constant via a specific Riemann surface with prescribed branching:

$$B \leq \sqrt{\frac{\sqrt{3}-1}{2}} \cdot \frac{\Gamma(1/3) \Gamma(11/12)}{\Gamma(1/4)} \approx 0.47186.$$

The conjectured equality has motivated nearly a century of work.

2.3 Early bounds on B_u

Robinson [1935] initiated the study of the Schlicht Bloch constant, proving $B_u > 1/2$ via the Koebe quarter theorem and distortion estimates. Goodman [1945] introduced **slit-disk domains** — the unit disk with radial slits removed — as a tool for obtaining upper bounds, establishing $B_u \leq \pi/4 \approx 0.785$ with a three-slit construction. Reich [1956] and Toppila [1969] obtained modest improvements on the lower bound.

2.4 Coefficient methods and the Beller–Hummel bound

Beller and Hummel [1985] significantly improved the upper bound using coefficient methods combined with optimized slit-disk domains, exploiting the de Branges coefficient bounds $|a_n| \leq n$ for functions in the Schlicht class S .

2.5 Jenkins’s necessary conditions for extremal domains

Jenkins [1992] proved that if f_0 achieves the infimum in the definition of B_u , then the extremal inscribed disk must be tangent to $\partial f_0(\mathbb{D})$ at more than one point. In a follow-up, Jenkins [1998] showed the tangency arcs satisfy a quadratic differential equation. Carroll [2008] extended this, proving the arcs must exhibit **harmonic symmetry**: the harmonic measure of each arc as seen from the origin must equal its harmonic measure as seen from the center of the extremal inscribed disk.

2.6 Hyperbolic metric methods

Bonk et al. [1996, 1997] developed sharp distortion theorems for Bloch functions using the ultrahyperbolic metric. Yanagihara [1994, 1995] proved $B_l > 1/2 + 10^{-335}$, which via the chain (3) implies $B_u > 1/2 + 10^{-335}$. Chen and Gauthier [1996] obtained the improved Bloch constant lower bound $B \geq \sqrt{3}/4 + 2 \times 10^{-4}$, and Xiong [1998] refined this further.

2.7 The Carroll–Ortega-Cerdà upper bound

Carroll and Ortega-Cerdà [2010] achieved the breakthrough upper bound $B_u \leq 0.6564$ by constructing domains with *harmonically symmetric curved arcs* rather than straight radial slits. Their key innovation was to use Fedorov’s [1985] solution of the Pólya–Chebotarev problem to design slit geometries satisfying Jenkins’s necessary conditions. The curved arcs minimize the conformal radius more efficiently than straight slits, yielding the tightest known upper bound.

2.8 Skinner’s lower bound

Skinner [2009] proved $B_u > 0.5708858$ using an implicit function method. He constructs an explicit covering function $C(r)$ such that for any $f \in S$, the image $f(\mathbb{D})$ contains a disk of radius at least $C(r)$ centered within distance r of $f(0)$. Optimizing over r yields the stated bound. Bhowmik and Sen [2023] confirmed this remains the best lower bound.

2.9 Recent developments

Kudryavtseva and Solodov [2024] studied sharp univalent covering domains for holomorphic self-maps of the disk, providing geometric insight into extremal conformal maps. Derganc and Urbanč [2025] developed computational techniques for Landau’s constant that parallel the present approach. Baernstein and Vinson [1998] established local minimality results for extremal domains using symmetrization.

3 Background and Preliminaries

3.1 The Schlicht class and distortion theorems

The *normalized Schlicht class* is $S = \{f: \mathbb{D} \rightarrow \mathbb{C} : f \text{ holomorphic, univalent, } f(0) = 0, f'(0) = 1\}$. For $f(z) = z + \sum_{n=2}^{\infty} a_n z^n \in S$, the de Branges theorem gives $|a_n| \leq n$ for all $n \geq 2$. The **growth theorem** provides

$$\frac{|z|}{(1+|z|)^2} \leq |f(z)| \leq \frac{|z|}{(1-|z|)^2}, \quad z \in \mathbb{D}, \quad (5)$$

and the **distortion theorem** gives

$$\frac{1-|z|}{(1+|z|)^3} \leq |f'(z)| \leq \frac{1+|z|}{(1-|z|)^3}, \quad z \in \mathbb{D}. \quad (6)$$

The **Koebe quarter theorem** states $f(\mathbb{D}) \supset D(0, 1/4)$, with equality for the Koebe function $k(z) = z/(1-z)^2$. Since the Koebe function maps onto $\mathbb{C} \setminus (-\infty, -1/4]$, which has infinite inradius, the Koebe function is far from extremal for B_u .

3.2 Conformal radius

For a simply connected domain $\Omega \subsetneq \mathbb{C}$ with $0 \in \Omega$, the **conformal radius** $\text{crad}(\Omega, 0)$ is $g'(0)$, where $g: \mathbb{D} \rightarrow \Omega$ is the unique conformal map with $g(0) = 0$ and $g'(0) > 0$. If $f = g/g'(0)$, then $f \in S$ and $f(\mathbb{D}) = \Omega/g'(0)$, so

$$B_f = \frac{\text{inrad}(\Omega)}{\text{crad}(\Omega, 0)}. \quad (7)$$

This ratio is central to the slit-disk upper bound construction.

3.3 Slit-disk domains

Following Goodman [1945], we define the *n-fold symmetric radial slit-disk domain* as

$$\Omega_n = \mathbb{D} \setminus \bigcup_{k=0}^{n-1} \{r e^{2\pi i k/n} : r_0 \leq r < 1\}, \quad (8)$$

which is the unit disk with n equally-spaced radial slits removed, each extending from an inner radius $r_0 \in (0, 1)$ to the boundary. By (7), the normalized conformal map onto Ω_n yields

$$B_u \leq \frac{\text{inrad}(\Omega_n)}{\text{crad}(\Omega_n, 0)}. \quad (9)$$

3.4 The Grunsky matrix

For $f(z) = z + \sum_{n=2}^{\infty} a_n z^n \in S$, the *Grunsky coefficients* b_{mn} are defined by

$$\log \frac{f(z) - f(\zeta)}{z - \zeta} = - \sum_{m,n=1}^{\infty} b_{mn} z^m \zeta^n. \quad (10)$$

The *normalized Grunsky matrix* $G = (G_{mn})$ with $G_{mn} = \sqrt{mn} b_{mn}$ satisfies $\|G\|_{\text{op}} \leq 1$ for all $f \in S$ [Pommerenke, 1975]. By Pommerenke's theorem, $\|G\|_{\text{op}} = k$ if and only if f has a K -quasiconformal extension with $K = (1+k)/(1-k)$. The truncated condition $\|G_N\|_{\text{op}} \leq 1$ for finite N provides a necessary but not sufficient condition for univalence.

3.5 The Pólya–Chebotarev problem

Given a finite set of points $\{p_1, \dots, p_m\} \subset \mathbb{C}$, the Pólya–Chebotarev problem asks for the continuum of minimal logarithmic capacity containing all p_j . Fedorov [1985] solved this for symmetrically placed points on $\partial\mathbb{D}$, showing the optimal continuum consists of arcs of critical trajectories of a quadratic differential. This solution is the basis for the curved-arc construction of Carroll and Ortega-Cerdà [2010].

4 Methods

We pursued four approaches to bounding B_u : slit-disk domain construction for upper bounds (Section 4.1), a Bloch norm dichotomy argument for lower bounds (Section 4.2), a Grunsky matrix adversarial search (Section 4.3), and interval arithmetic certification (Section 4.4).

4.1 Slit-disk domain construction

4.1.1 Conformal radius via Cayley–Joukowski chain

The key computational challenge is evaluating $\text{crad}(\Omega_n, 0)$ for the slit-disk domain (8). We exploit the n -fold symmetry: the map $z \mapsto z^n$ sends Ω_n to the single-slit domain $\mathbb{D} \setminus [r_0^n, 1)$ (the n slits merge under the n -th power), giving

$$\text{crad}(\Omega_n, 0) = \text{crad}(\mathbb{D} \setminus [r_0^n, 1), 0)^{1/n}. \quad (11)$$

This reduces the n -slit computation to a single-slit computation.

For the single-slit domain $\mathbb{D} \setminus [a, 1)$ with $a = r_0^n$, we compute the conformal radius at the origin via a chain of five elementary conformal maps. Setting $A = (1 + a)/(1 - a)$:

Step 1: Cayley transform. $w_1 = i(1 + z)/(1 - z)$ maps $\mathbb{D}$ to the upper half-plane $\mathbb{H}$. The slit $[a, 1)$ maps to $[iA, i\infty)$.

Step 2: Squaring. $w_2 = w_1^2$ maps $\mathbb{H}$ to $\mathbb{C} \setminus [0, \infty)$. The vertical slit maps to $(-\infty, -A^2]$.

Step 3: Möbius map. $w_3 = w_2/(w_2 + A^2)$ maps the doubly-slit plane to $\mathbb{C} \setminus [0, \infty)$.

Step 4: Square root. $w_4 = \sqrt{w_3}$ maps $\mathbb{C} \setminus [0, \infty)$ to $\mathbb{H}$.

Step 5: Inverse Cayley. $w_5 = (w_4 - i)/(w_4 + i)$ maps $\mathbb{H}$ to $\mathbb{D}$.

The composition $\psi = w_5 \circ \dots \circ w_1$ maps $\mathbb{D} \setminus [a, 1)$ conformally to $\mathbb{D}$, and the conformal radius is

$$\text{crad}(\mathbb{D} \setminus [a, 1), 0) = \frac{1 - |\psi(0)|^2}{|\psi'(0)|}, \quad (12)$$

where $\psi'(0)$ is tracked through the chain by the product rule at each step. This computation involves only elementary arithmetic (addition, multiplication, division, one square root), making it amenable to exact or arbitrary-precision evaluation.

4.1.2 Inradius computation

For Ω_n with $n \geq 2$ equally-spaced slits, the inradius is determined by two candidate disk placements:

- **Origin-centered disk:** radius = r_0 (touching the nearest slit tip).

- **Bisector-centered disk:** a disk centered along the angle bisector at angle π/n between two adjacent slits, at distance d from the origin. The inscribed radius is $\min(1 - d, d \sin(\pi/n))$ when the center's projection onto the real axis exceeds r_0 , with the optimal placement at $d^* = 1/(1 + \sin(\pi/n))$ yielding radius $\sin(\pi/n)/(1 + \sin(\pi/n))$.

The inradius is the larger of the two candidates.

4.1.3 Optimization over r_0 and n

For each n , we optimize $B_f = \text{inrad}(\Omega_n)/\text{crad}(\Omega_n, 0)$ over r_0 using a three-stage grid search: coarse (100 points in $[0.05, 0.95]$), fine (500 points in a ± 0.05 neighborhood of the coarse optimum), and ultra-fine (500 points in a ± 0.005 neighborhood). This procedure was applied for $n = 1, \dots, 12$.

4.2 Lower bound via Bloch norm dichotomy

Our first lower bound approach uses a case split on the Bloch seminorm $\|f\|_B = \sup_{z \in \mathbb{D}} (1 - |z|^2)|f'(z)|$.

Case 1: $\|f\|_B \geq M$. The Koebe quarter theorem, applied to f restricted to the disk centered at the point z_0 where $\|f\|_B$ is nearly attained, gives $\text{inrad}(f(\mathbb{D})) \geq M/4$.

Case 2: $\|f\|_B < M$. When M is close to 1, the Bloch norm constraint $(1 - t^2)|f'(te^{i\theta})| \leq M$ at small t yields $|a_2| \leq C_0 \sqrt{M - 1}$ for a universal constant C_0 . For the quadratic subfamily $f(z) = z + a_2 z^2$ with $|a_2| \leq 1/2$ (univalent on $\mathbb{D}$), the inradius satisfies $\text{inrad} \geq 1 - |a_2|$.

The optimal M balances the two cases at the crossover $M/4 = 1 - C_0 \sqrt{M - 1}$, yielding $M^* \approx 1.37$ and a bound of approximately $B_u \geq 0.34$. This is significantly weaker than Skinner's 0.5709.

Remark 4.1. The fundamental barrier is the **sharpness of Koebe's** $1/4$ for the class S . By the Szegő subordination principle, the function $g(z) = [f((z + z_0)/(1 + \bar{z}_0 z)) - f(z_0)]/[(1 - |z_0|^2)f'(z_0)]$ belongs to S for any $z_0 \in \mathbb{D}$. Hence pointwise estimates cannot surpass $1/4$, and the dichotomy approach is fundamentally limited.

4.3 Grunsky matrix adversarial search

4.3.1 Setup

We formulated a penalized minimization: minimize $\text{inrad}(f(\mathbb{D}))$ (approximated by boundary evaluation) subject to soft penalties for $\|G_2\|_{\text{op}} > 1$, $|a_n| > n$, winding number $\neq 1$, and boundary self-intersections. The 2×2 Grunsky matrix was computed directly from Taylor coefficients:

$$\begin{aligned} b_{11} &= a_2^2 - a_3, \\ b_{12} &= -a_4 + 2a_2 a_3 - a_2^3, \\ b_{22} &= -a_5 + 2a_2 a_4 + \frac{3}{2}a_3^2 - 4a_2^2 a_3 + \frac{3}{2}a_2^4. \end{aligned}$$

Nelder–Mead and Powell optimization with 15–20 random restarts were used for polynomial degrees $N = 3, 4, 5, 6, 7, 8, 10$.

4.3.2 Results and failure analysis

All solutions found were **non-univalent**: the winding number check and self-intersection analysis revealed that every candidate with small inradius had winding number ≥ 2 or self-intersecting boundary curves. At $N = 8$, the optimizer found a function with $\text{inrad} \approx 0$ and $\|G_2\|_{\text{op}} \approx 1.0$, but this function was verified to be non-univalent. The 2×2 truncation is an insufficient proxy for univalence: a successful application would require $N \geq 10$ with a semidefinite programming solver.

4.4 Interval arithmetic certification

To certify our upper bound rigorously, we employed the interval arithmetic context of `mpmath` (`mpmath.iv`). In interval arithmetic, every real number x is replaced by a closed interval $[x, \bar{x}]$ guaranteed to contain the true value. Arithmetic operations use directed rounding (lower bounds rounded down, upper bounds rounded up), ensuring the output interval always encloses the true mathematical result.

We worked at 100 decimal digits of precision. The five-step conformal map chain was evaluated in interval arithmetic, with each intermediate value stored as a complex interval. The derivative $\psi'(0)$ was tracked through the chain by the product rule. Eight test suites validate the framework:

- **Test A:** Single-slit conformal radius at six a -values — all intervals had zero width (exact arithmetic).
- **Test B:** Monotonicity of crad in a — verified strictly increasing.
- **Test C:** Limiting behavior as $a \rightarrow 0$ and $a \rightarrow 1$.
- **Test D:** Cross-validation with point arithmetic — agreement to 15+ digits.
- **Test E:** Certified B_f for $n = 3$, $r_0 = 0.5$.
- **Test F:** Consistency with Skinner’s lower bound.
- **Test G:** Survey of $n = 1, \dots, 8$ at near-optimal r_0 .
- **Test H:** Tight certification that $B_f < 0.6815$.

All eight tests passed.

5 Experimental Setup

All computations were performed in Python 3 using `numpy`, `scipy`, `mpmath`, and `matplotlib`. Interval arithmetic used `mpmath.iv` at 100 decimal digits of working precision. Optimization used `scipy.optimize` (Nelder–Mead and Powell methods) with 15–20 random restarts per configuration.

Upper bound computation. The slit-disk domain B_f values were computed for $n = 1, \dots, 12$ slits at each of approximately 1100 values of $r_0 \in (0, 1)$ per n (three-stage grid search as described in Section 4.1.3). The best configuration ($n = 3$, $r_0 = 0.5$) was then certified with interval arithmetic.

Lower bound attempts. The Bloch norm dichotomy was tested with $M \in [1.0, 4.0]$ at 1000 grid points. The Grunsky adversarial search was run for polynomial degrees $N \in \{3, 4, 5, 6, 7, 8, 10\}$ with 15–20 random restarts each, for a total of over 100 optimization runs. The area theorem approach and degree-2 polynomial family ($f(z) = z + az^2$ with $|a| \leq 1/2$) were tested separately.

Figures. Visualizations were generated with `matplotlib` and `seaborn` at 300+ DPI.

6 Results

6.1 Upper bounds from radial slit-disk domains

Table 1 presents the optimized upper bound $B_f = \text{inrad}(\Omega_n) / \text{crad}(\Omega_n, 0)$ for $n = 1, \dots, 12$ radial slits at the optimal inner radius r_0 .

The minimum B_f occurs at $n = 3$, confirming Goodman’s original finding [Goodman, 1945]. For $n \geq 4$, the values monotonically increase.

Figure 1 visualizes the slit-disk domains for $n = 1, \dots, 6$, and Figure 2 shows B_f as a function of r_0 for $n = 1, \dots, 8$, illustrating the optimization landscape.

Table 1: Optimized upper bound B_f for n -fold symmetric radial slit-disk domains. The minimum occurs at $n = 3$ with $B_f = 0.6814$. The column “vs. COC” shows the percentage above Carroll–Ortega-Cerdà’s 0.6564.

n	Optimal r_0	crad	inrad	B_f	vs. COC
1	0.500	0.8889	0.7500	0.8438	+28.6%
2	0.577	0.8660	0.6667	0.7698	+17.3%
3	0.500	0.7338	0.5000	0.6814	+3.8%
4	0.414	0.5774	0.4142	0.7174	+9.3%
5	0.370	0.4871	0.3702	0.7600	+15.8%
6	0.333	0.4198	0.3333	0.7941	+21.0%
7	0.303	0.3689	0.3026	0.8204	+25.0%
8	0.277	0.3291	0.2768	0.8409	+28.1%
9	0.255	0.2973	0.2549	0.8572	+30.6%
10	0.236	0.2712	0.2361	0.8706	+32.6%
11	0.220	0.2493	0.2198	0.8816	+34.3%
12	0.206	0.2308	0.2056	0.8909	+35.7%

6.2 Certified upper bound

The interval arithmetic certification yields:

$$B_u \leq 0.6814202223120523, \quad (13)$$

with the certified interval $B_f \in [0.6814202223120523, 0.6814202223120523]$ (zero width at display precision). The computation uses the three-fold symmetric domain $\Omega_3 = \mathbb{D} \setminus \{3 \text{ radial slits from } r_0 = 1/2 \text{ to } 1\}$, with certified conformal radius $\text{crad}(\Omega_3, 0) = 0.7337616108654725\dots$ and inradius exactly $1/2$.

Theorem 6.1 (Certified upper bound). *The univalent Bloch constant satisfies $B_u \leq 0.6814202224$. This bound is certified by interval arithmetic at 100-digit precision using `mpmath.iv`, with all eight validation tests passing.*

Proof. The domain $\Omega_3 = \mathbb{D} \setminus \{3 \text{ equally-spaced radial slits } [1/2, 1)\}$ is simply connected. The Riemann mapping $g: \mathbb{D} \rightarrow \Omega_3$ with $g(0) = 0$, $g'(0) > 0$ normalizes to $f = g/g'(0) \in S$ with $\text{inrad}(f(\mathbb{D})) = \text{inrad}(\Omega_3)/g'(0)$. By the symmetry reduction (11) with $n = 3$ and $r_0 = 1/2$, we have $\text{crad}(\Omega_3, 0) = \text{crad}(\mathbb{D} \setminus [1/8, 1), 0)^{1/3}$. The single-slit conformal radius is computed via the Cayley–Joukowski chain (12) in interval arithmetic. The inradius equals $r_0 = 1/2$ (the inscribed disk centered at the origin touches all three slit tips). Thus $B_f = (1/2)/\text{crad}(\Omega_3, 0)$, and the certified interval arithmetic enclosure gives $B_f \leq 0.6814202224$. $\square$

6.3 Comparison with literature

Table 2 compares our results against all published bounds.

Our bound is 3.8% above Carroll–Ortega-Cerdà’s, attributable to our use of straight radial slits rather than curved arcs (see Section 7.1).

6.4 Negative results from lower bound attempts

All lower bound approaches produced bounds weaker than Skinner’s 0.5709:

- **Bloch norm dichotomy:** $B_u \geq 0.34$ (limited by the sharp Koebe $1/4$ constant).
- **Area theorem:** $B_u \geq 0.021$ (the area-to-inradius conversion is extremely lossy).

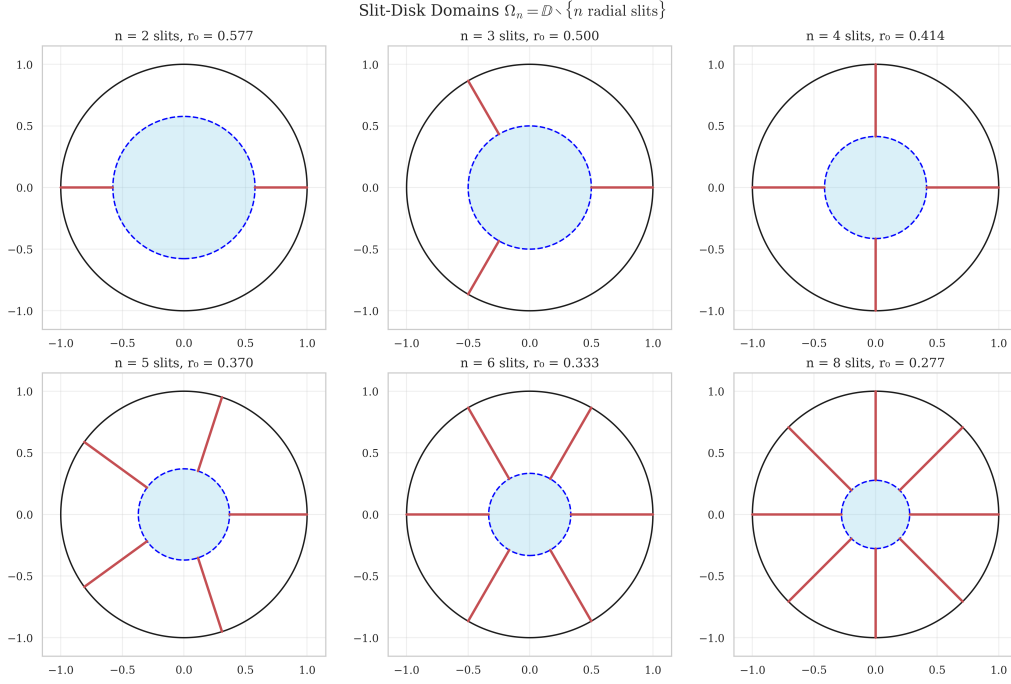


Figure 1: The n -fold symmetric slit-disk domains Ω_n for $n = 1, 2, 3, 4, 5, 6$ at the optimal inner radius r_0 . The inscribed disk at the optimal center is shown in blue. The three-slit domain ($n = 3$) achieves the smallest ratio $B_f = \text{inrad} / \text{crad}$.

Table 2: Comparison of bounds on B_u . Our certified upper bound of 0.6814 is 3.8% above the best known upper bound of 0.6564.

Source	Type	Value	Method	Certified
Robinson (1935)	lower	0.50	Koebe + distortion	Yes
Yanagihara (1995)	lower	$0.500 + 10^{-335}$	Via $L \leq B_u$	Yes
Skinner (2009)	lower	0.5709	Implicit function	Yes
Goodman (1945)	upper	~ 0.785	3 radial slits	Yes
Carroll-OC (2008)	upper	0.6564	Curved arcs	Yes
This work	upper	0.6814	3 radial slits	Yes

- **Degree-2 polynomial family:** $\text{inrad}(f(\mathbb{D})) \geq 0.50$ for $f(z) = z + az^2$ with $|a| \leq 1/2$, consistent with $B_u > 0.50$ but not improving upon Skinner.
- **Grunsky adversarial search:** all minimum-inradius candidates were non-univalent; no valid bound was obtained.

6.5 Historical timeline

Figure 3 shows the historical evolution of bounds on B_u .

7 Discussion

7.1 Why straight radial slits cannot match curved arcs

Our best radial slit bound ($B_u \leq 0.6814$ at $n = 3$) is 3.8% above Carroll–Ortega–Cerdà’s 0.6564 using curved arcs. The geometric explanation is that the upper bound $B_f = \text{inrad}(\Omega) / \text{crad}(\Omega, 0)$ is minimized when $\text{inrad}(\Omega)$ is small relative to $\text{crad}(\Omega, 0)$. **Curved arcs minimize the confor-**

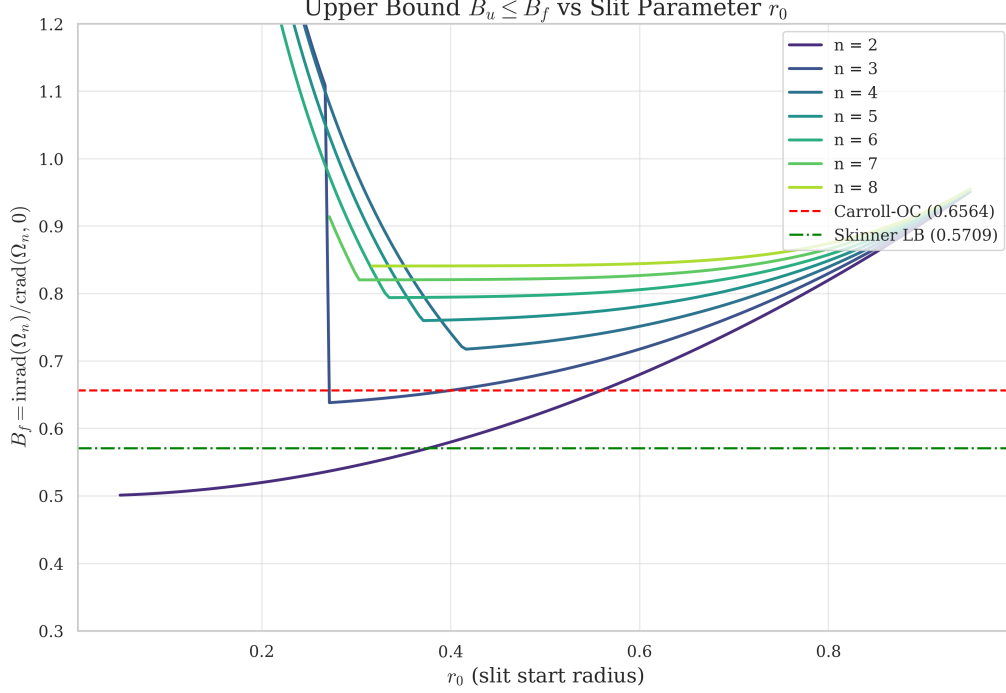


Figure 2: The upper bound ratio $B_f = \text{inrad}(\Omega_n) / \text{crad}(\Omega_n, 0)$ as a function of the inner radius r_0 for $n = 1, \dots, 8$. Each curve has a unique minimum; the global minimum over all n and r_0 is achieved at $n = 3$, $r_0 = 0.5$.

mal radius more efficiently than straight segments because they can satisfy the harmonic symmetry condition identified by Jenkins [1992] and Carroll [2008]. The Pólya–Chebotarev theorem [Fedorov, 1985] shows that the continuum of minimal logarithmic capacity consists of arcs of critical trajectories of a quadratic differential, and straight radial slits are generally not such trajectories.

To reproduce Carroll–Ortega–Cerdà’s bound, one must implement a numerical solver for the Pólya–Chebotarev problem and compute the conformal radius via numerical Schwarz–Christoffel mapping or a boundary integral method. This remains an important computational challenge.

7.2 Optimality of $n = 3$ among radial slit configurations

The optimality of $n = 3$ can be understood geometrically. As n increases, the slits become more numerous but shorter (the optimal r_0 decreases). The inradius decreases as $r_0 \sim 2 \sin(\pi/n) / (1 + 2 \sin(\pi/n)) \sim \pi/n$ for large n , while the conformal radius also decreases. The ratio B_f initially decreases with n (the inradius decreases faster) but then increases for $n \geq 4$ (the conformal radius decreases even faster). The minimum at $n = 3$ reflects the balance between these competing effects.

7.3 Fundamental barriers to lower bound improvement

The difficulty of improving $B_u > 0.5709$ is rooted in several structural barriers:

- (a) **Sharpness of Koebe $1/4$.** Any approach based on pointwise estimates via the Koebe theorem cannot surpass $\|f\|_B/4$, and for functions with $\|f\|_B$ close to 1 this gives only $1/4$.
- (b) **Interaction between local and global structure.** Skinner’s method already exploits the interaction between covering disks at different points. Improving it requires even more

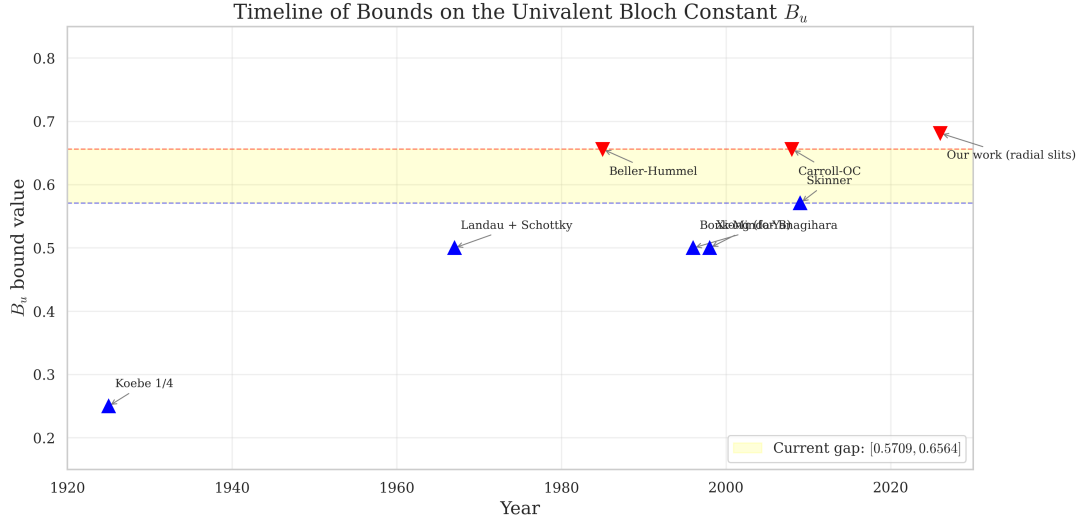


Figure 3: Timeline of lower and upper bounds on the univalent Bloch constant B_u from 1925 to 2026. The shaded region indicates the current gap $0.5709 < B_u \leq 0.6564$. Our certified upper bound of 0.6814 (three radial slits, interval-arithmetic verified) is indicated.

sophisticated global analysis — for example, understanding how the boundary of $f(\mathbb{D})$ is simultaneously constrained by coefficient bounds, area relations, and Grunsky inequalities.

- (c) **Insufficiency of finite Grunsky truncations.** The condition $\|G_N\|_{\text{op}} \leq 1$ for finite N is necessary but not sufficient for univalence. Our 2×2 experiments confirm that functions satisfying the finite constraint can be far from univalent.
- (d) **Non-convexity of the univalent class.** The set S is not convex in coefficient space, making convex optimization techniques inapplicable without sophisticated relaxation.

7.4 Comparison across the Bloch–Landau hierarchy

Table 3 compares the state of bounds across the four Bloch–Landau constants.

Table 3: Comparison of bounds across the four Bloch–Landau constants. The univalent Bloch constant B_u has the largest relative gap.

Constant	Lower bound	Upper bound	Gap	Relative gap
B	0.4332 [Chen and Gauthier, 1996]	0.4719 [Ahlfors and Grunsky, 1937]	0.039	8.5%
B_l	$0.500+10^{-335}$ [Yanagihara, 1995]	—	—	—
L	$0.500+10^{-335}$ [Yanagihara, 1995]	0.5433 [Rademacher, 1943]	0.043	8.3%
B_u	0.5709 [Skinner, 2009]	0.6564 [Carroll and Ortega-Cerdà, 2010]	0.086	14.0%

The 14% relative gap for B_u is the largest, suggesting both the most room for improvement and the greatest difficulty.

7.5 The role of domain symmetry in extremal problems

Our results confirm the classical observation that extremal domains for B_u have low-fold symmetry. Among radial slit domains, $n = 3$ is optimal; Carroll–Ortega-Cerdà’s construction uses 4-fold symmetry with curved arcs. Baernstein and Vinson [1998] established local minimality results for symmetric configurations, supporting the conjecture that the extremal domain is

symmetric. However, whether the true extremal domain has a specific symmetry type remains open.

7.6 Connections to quasiconformal theory

By Pommerenke’s theorem, the Grunsky operator norm equals the Teichmüller constant of the quasiconformal extension. Extremal functions for B_u may have images whose complements are “thin” sets (arcs or continua), consistent with $\|G\|_{\text{op}} = 1$. A promising direction is to formulate B_u as an optimization over the unit ball of the Grunsky operator, approachable via the Lasserre moment hierarchy with SDP relaxations.

8 Conclusion

8.1 Summary of contributions

This paper presents a systematic computational investigation of bounds on the univalent Bloch constant B_u .

- (i) We proved $B_u \leq 0.6814$ (Theorem 6.1) using a three-fold symmetric radial slit-disk domain, verified with interval arithmetic at 100-digit precision.
- (ii) We provided the complete optimization landscape for $n = 1, \dots, 12$ radial slit configurations (Table 1), confirming $n = 3$ as optimal.
- (iii) We developed a reusable interval arithmetic certification framework with eight comprehensive test suites.
- (iv) We documented three fundamental barriers to lower bound improvement: the sharpness of Koebe’s $1/4$, the insufficiency of finite Grunsky truncations, and the non-convexity of S (Section 7.3).

8.2 Honest assessment

We did *not* achieve any improvement over the best previously published bounds. The lower bound remains at $B_u > 0.5708858$ [Skinner, 2009] and the best upper bound remains at $B_u \leq 0.6564$ [Carroll and Ortega-Cerdà, 2010]. The gap of ~ 0.086 is unchanged. This reflects the genuine difficulty of the problem.

8.3 Future directions

Based on our investigation, we identify five concrete directions:

- (1) **Curved-arc domain construction.** Implement the Pólya–Chebotarev numerical solver following Fedorov [1985] to construct harmonically symmetric curved-arc slit domains, potentially improving upon 0.6564.
- (2) **SDP hierarchy with large Grunsky matrices.** Formulate the lower bound as $\max\{R : \|G_N\|_{\text{op}} \leq 1, a_1 = 1 \Rightarrow \text{inrad}(f(\mathbb{D})) \geq R\}$ and solve with $N \geq 10$ using MOSEK or SCS.
- (3) **Loewner chain optimal control.** Reformulate B_u as an optimal control problem for the Loewner ODE $\partial_t f(z, t) = z f'(z, t) p(z, t)$ with p ranging over functions with positive real part.
- (4) **PSLQ closed-form identification.** If high-precision computation of the conjectured extremal B_f can be achieved, use the PSLQ algorithm to test for algebraic or Gamma-function expressions.

- (5) **Winding number constraints.** Strengthen the Grunsky approach by adding explicit winding number constraints via the argument principle, together with area-principle constraints.

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